

AI Applications in Adolescent Mental Health (2015–2024): A Systematic Literature Review

1. Introduction

Mental health is an important problem for teenagers. During adolescence, young people experience many changes in their body, study, family and social life. Because of these changes, some teenagers may have problems such as depression, anxiety, stress and self-harm. If these problems are not noticed early, they can have a bad effect on their future life.

In recent years, Artificial Intelligence (AI) has become more popular in healthcare. AI can use computers to find patterns from a large amount of information. In mental health, AI can be used to find possible mental health problems, predict risks and give some basic support to patients.

There are also more and more digital mental health applications. For example, some applications use chatbots to talk with users. Other systems can analyse text from social media or information from questionnaires. These technologies may be useful for teenagers because many young people use smartphones and the Internet every day.

However, using AI for mental health also has some problems. Mental health information is private and sensitive. If this information is collected by an AI system, there can be privacy problems. Also, AI can make mistakes. It may say that someone has a mental health problem when they do not have one, or it may not find a real problem.

This literature review looks at how AI has been used in adolescent mental health between **2015 and 2024**. It also looks at the benefits, problems and future research of using AI in this area.

2. Research Question

The main research question of this review is:

How has AI been used in adolescent mental health between 2015 and 2024?

There are also three smaller questions:

1. What AI technologies are used for adolescent mental health?
 2. What mental health problems can AI help to find or support?
 3. What are the main benefits and problems of using AI for teenagers?
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3. Methodology

This study uses a **Systematic Literature Review (SLR)** method. A systematic literature review means finding research papers about one topic in a planned way, and then reviewing the results.

The research focused on studies published between **2015 and 2024**. The main databases that can be used for this topic are PubMed, Scopus, Web of Science and PsycINFO.

The search words included:

- Artificial Intelligence
- AI
- Machine Learning
- Natural Language Processing
- Chatbot
- Adolescent
- Teenager
- Youth
- Mental Health
- Depression
- Anxiety
- Suicide
- Self-harm

A possible search sentence is:

("artificial intelligence" OR "machine learning" OR chatbot) AND (adolescent OR teenager OR youth) AND ("mental health" OR depression OR anxiety OR suicide)

The papers were selected based on the topic and the age group. Papers that were only about adults were not included.

4. Inclusion and Exclusion Criteria

Some rules were used to decide which papers should be included.

Inclusion criteria

The paper should:

- Be published between 2015 and 2024.
- Be related to artificial intelligence.
- Be related to mental health.
- Include adolescents or young people.
- Be written in English.
- Be a research paper or a useful review paper.

Exclusion criteria

The paper was not included if:

- It was only about adults.
- It was not related to mental health.
- It did not use AI or machine learning.

- It was only a general article about AI.
- It was not written in English.
- It was a duplicate paper.

The study selection can be shown by a PRISMA diagram. First, papers are found from different databases. Then duplicate papers are removed. After that, the title and abstract are checked. The remaining papers are read in full, and the papers that meet the criteria are included in the final review.

5. Results

After looking at the research about AI and adolescent mental health, several main applications can be found.

The main uses are:

1. Finding mental health problems
2. Predicting mental health risks
3. Chatbots and digital support
4. Monitoring mental health

5.1 Finding Mental Health Problems

One of the main uses of AI is finding possible mental health problems.

Machine learning can look at a lot of information and find patterns. For example, researchers can use questionnaires, medical information, online text and other information to train a computer model.

The AI model may then try to identify people who have a higher chance of having depression or anxiety.

This can be useful because some teenagers may not tell their parents or teachers when they have mental health problems. If an AI system can find possible problems earlier, the teenager may get help earlier.

However, AI does not always give the correct answer. A computer model can work well in one research study but may not work as well with another group of people.

5.2 Predicting Suicide and Self-Harm

Another important area is suicide and self-harm.

Some researchers use machine learning to find teenagers who may have a higher risk of suicidal thoughts or self-harm. The AI can look at different types of information at the same time.

For example, it may use previous mental health information, questionnaires, behaviour and online information.

Social media is also studied in this area. Some researchers use Natural Language Processing (NLP) to analyse the words people use online. Certain words or sentences may be related to depression or suicidal thoughts.

This can be useful for early prevention. But it also creates a serious privacy problem. People may not want their social media posts to be analysed by an AI system.

Also, having some words related to suicide does not always mean that a person is going to hurt themselves. Therefore, AI should not make the final decision by itself.

5.3 Chatbots and Digital Mental Health Support

Chatbots are another popular use of AI.

A chatbot is a computer program that can communicate with people using text. Some mental health chatbots are designed to give basic psychological support.

One famous example is **Woebot**. Fitzpatrick et al. (2017) studied a chatbot that used ideas from Cognitive Behavioural Therapy (CBT). The study found that people using the chatbot had a reduction in depression symptoms.

However, this study mainly included young adults, not younger teenagers. Therefore, the result cannot be directly applied to all adolescents.

Chatbots can have some advantages. They can be used at any time and the user does not need to make an appointment. Some young people may also feel more comfortable talking to a computer.

But a chatbot is not the same as a real psychologist. A computer may not understand a complicated personal situation. If someone has a serious mental health problem, they should still get help from a qualified professional.

5.4 Monitoring Mental Health

AI can also be used to monitor mental health.

For example, an application can collect information about mood, sleep, activity or answers from questionnaires. AI can then look at this information and try to find changes.

This could help doctors or other professionals understand a person's mental health over a longer time.

For teenagers, this could be useful because their mood can change quite quickly. An application may notice some changes before the person talks to a doctor.

However, collecting this information also creates privacy concerns. People need to know what information is being collected and who can see it.

6. Benefits of AI in Adolescent Mental Health

There are several possible benefits of AI.

6.1 Easy to Access

AI applications can be available on a phone or computer. Teenagers can use them from home and they may not need to travel to a hospital or clinic.

This can be useful for people who live far away from mental health services.

6.2 Early Detection

AI may help find mental health problems earlier. Early detection is important because getting help earlier may stop the problem from becoming worse.

6.3 Less Stigma

Some teenagers may feel embarrassed about talking to a psychologist. Talking with a computer may feel easier for some people.

However, this does not mean everyone will prefer AI.

6.4 Can Process Large Amounts of Data

AI can process a large amount of information much faster than a person. This is useful when researchers need to look at many people's data.

7. Problems and Risks

Although AI has benefits, there are also many problems.

7.1 Privacy

Privacy is probably one of the biggest problems.

Mental health information is very personal. If an AI application collects information about depression, anxiety or suicide, this information must be protected.

There is also a problem when social media data is used. People may not know that their posts are being used for mental health research.

7.2 AI Can Make Mistakes

AI is not perfect.

For example, it may think someone has depression when they do not. It can also miss someone who really needs help.

This can be dangerous when AI is used for serious problems such as suicide risk.

7.3 Different People May Get Different Results

AI models are trained using data. If the data mainly comes from one country or one type of person, the model may not work well for other people.

For example, teenagers from different cultures may use different words to describe their feelings. An AI model may not understand all of these differences.

7.4 AI Cannot Fully Replace Humans

Mental health is not only about data.

A psychologist can listen to a person and understand their family situation, emotions and personal experience. AI cannot fully do this.

Therefore, AI should mainly be used as a support tool, not as a replacement for human professionals.

8. Discussion

From the studies reviewed, AI has a lot of potential in adolescent mental health. The most common use seems to be finding and predicting mental health problems.

Machine learning is useful because it can find patterns from large amounts of information. This may help with early detection of depression, anxiety and suicide risk.

Chatbots are also interesting because they can give support at any time. They may be especially useful for people who do not want to talk to a professional at first.

However, there is still not enough evidence to say that AI is better than normal mental health services. Many studies are small and use different methods. Some studies also focus on young adults instead of teenagers.

Another problem is that technology is developing very quickly. An AI system that was studied several years ago may already be different from current AI systems.

Because of this, more research is needed before AI can be widely used for adolescent mental health.

9. Research Gaps

There are several gaps in the current research.

First, there are not enough **long-term studies**. Some research only follows people for a short time. More studies should check whether AI support is still useful after several months or years.

Second, more research should focus specifically on **teenagers**. Some studies include people from 18 to 25 years old, but this is not exactly the same as younger adolescents.

Third, more countries and cultures should be included. Most research cannot represent every teenager in the world.

Fourth, researchers need to study **privacy and safety** more carefully. This is especially important because mental health information is sensitive.

Finally, there should be more research about **new generative AI systems**. ChatGPT-like systems became very popular around the end of the review period. They can communicate more naturally than older chatbots, but they can also give wrong information. More research is needed before they are used for serious mental health support.

10. Limitations of This Review

This review also has some limitations.

First, only English language papers were considered. Therefore, some useful studies in other languages may be missing.

Second, the review period was limited to 2015–2024. New studies after 2024 were not included.

Third, different studies used different AI technologies and different ways to measure mental health. Because of this, it is difficult to directly compare all the results.

Also, AI is changing very fast. Some findings from older studies may not be completely suitable for today's AI technology.

11. Conclusion

This systematic literature review looked at the use of AI in adolescent mental health from 2015 to 2024.

The review shows that AI is mainly used for finding mental health problems, predicting risks, providing digital support and monitoring people's mental health.

AI has some important benefits. It can be easy to access, process a lot of information and possibly help find problems earlier. Chatbots may also give young people a simple way to get some basic support.

However, there are also important problems. AI can make mistakes, personal information can be leaked, and AI may not work equally well for all groups of people. More importantly, AI cannot fully understand human emotions and personal situations like a trained mental health professional can.

Therefore, AI should not replace psychologists or doctors. It should be used together with human support. More research is needed, especially research with teenagers, long-term studies and studies about privacy and safety.

Overall, AI has a good potential in adolescent mental health, but it is still a developing area. In the future, if the technology can become safer and more accurate, it may become a useful part of mental health services for young people.

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